

SAILENDRA AKASH BONAGIRI

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EDUCATION

University of California, Davis

PhD in Computer Science (Research focus - Applied ML, Multi-AI Agent Evals)

Sep 2022 - Present

Davis, California

- Coursework: Machine Learning, Advanced Deep Learning, NLP, Web & Data Analytics, Distributed Databases

Birla Institute of Technology and Science, Pilani (BITS PILANI)

Bachelor of Engineering in Electronics and Communication

Aug 2017 - May 2021

Hyderabad, India

- Coursework: Data Structures & Algorithms, OOP, Operating Systems, Statistics, AI, HCI, Stochastic Optimization

PUBLICATIONS

- “**Safer Social Platforms**: Scalable Performant Few-Shot Harm Moderation Using LLMs” submitted to **ACL 2025** [↗](#)
- “**NLP4Gov**: A Comprehensive Search & Retrieval Library for Computational Policy Analysis.” In **CHI, 2024** [↗](#)
- “**Aletheia**: Fake News Detection System for Hindi.” In **ACM IKDD CODS COMAD, 2022** [↗](#) (Best Paper Award)
- “**Learnings** from Attempt to Build NLP Fake News Classification system for Hindi.” In **ACM SIGCAS, 2021** [↗](#)

EXPERIENCE

Machine Learning Researcher – PRISM AI Group [↗](#)

Sep 2024 – Present

Technologies Used: Python, LangChain, LangGraph, RAG, OpenAI Evals, HuggingFace, MCP

California, USA

- **Applied Machine Learning**
 - * Built a **Medical AI Agent** using real human data and user prompts for accurate, explainable diagnosis generation.
 - * Developed deep **spatiotemporal models** for traffic prediction and anomaly detection using large-scale sensor data.
 - * Modeled social media discourse to predict engagement & used **Perspective API** for toxicity and sentiment scoring.
- **AI Agents and Evaluations**
 - * Designed systematic **Agent Evaluation Pipelines** using **Evals** to assess reasoning, robustness, and reliability.
 - * Created **ResearchQuest AI** to automate research retrieval, summarization & analysis, cutting review time by **70%**.
 - * Orchestrated **Self-Verification Agents** for post training, alignment and reasoning transparency.
- **Benchmarking**
 - * Curated standardized **Benchmarks for AI Agent Evaluation** focused on reasoning depth and retrieval efficiency.
 - * Integrated **TerminalBench** and **AutoEval** [↗](#) frameworks to quantify agent task success and context transfer.
 - * Collaborated on reproducible testing standards for **Agentic Systems and Evals Research**.

Applied Machine Learning Researcher - C2 AI Lab [↗](#)

Sep 2022 - Sep 2024

Technologies Used: Pytorch, RAG, LangChain, Transformers, Gen AI LLMs, GCP

California, USA

- Investigated ICL and **Instruction fine-tuning** of open & closed source LLMs & Multimodal models in few-shot learning for harm detection in video platforms like Youtube etc., achieving **30%** higher accuracy compared to **SOTA LLMs**
- Engineered **NLP4GOV** [↗](#), an advanced open-source **NLP based search and retrieval library** for long document retrieval reducing document search time by **60%** and enhancing precision by **25%**

Machine Learning Engineer - PepsiCo [↗](#)

Jun 2021 - Sep 2022

Technologies Used: TensorFlow, FastAPI, Streamlit, Azure, Docker, Kubernetes, Hadoop

Hyderabad, India

- Led the development of **Precision-Agro SaaS ML** application to evaluate potato plant health, addressing inefficiencies in manual assessment through multimodal text and image recognition, resulting in a **35%** increase in crop yield
- Built a **data-driven recommendation system** using **ML** models, enhancing overall Gross Margin by **15%**
- Developed an application for field workers to input potato leaf images, reducing Time to Conversion by **40%**

Machine Learning Engineer - Innoviti Payment Solutions [↗](#)

Jul 2020 - Dec 2020

Technologies Used: Python, EDA, Predictive Models, Recommendation Systems, ML

Bengaluru, India

- Utilized Exploratory Data Analysis (**EDA**) for feature engineering, **predictive modeling** and **time series forecasting** to estimate future merchant trends, optimizing merchant selection and improving retrieval algorithms efficiency
- Leveraged ML insights on merchant preferences, enabling tailored offerings and increasing **customer retention** by **35%**

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, Javascript, R, HTML

Technologies: Git, Linux, FastAPI, AI Agent Builder, AWS, Elasticsearch, Kubernetes, Docker, Swagger, Postman

SWE, ML Tools: Tensorflow, Pytorch, PySpark, OpenCV, NLTK, RAG, LangChain, HuggingFace, LLMs, Transformers, Spark, Scikit-learn, numpy, pandas, seaborn, matplotlib, Github actions - CI/CD, Flask, React, RESTful API, PostgreSQL